CHAPTER 1

Solutions

Expressing Concentration of Solutions

1. The amount of glucose required to prepare 250 mL of $\frac{M}{20}$ aqueous solution is:
(Molar mass of glucose: 180 g mol^{-1}) *(2024 Re)*
a. 2.25 g b. 4.5 g c. 0.44 g d. 1.125 g

2. In one molal solution that contains 0.5 mole of a solute, there is *(2022)*
a. 1000 g of solvent b. 500 mL of solvent
c. 500 g of solvent d. 100 mL of solvent

3. Which of the following is dependent on temperature? *(2017-Delhi)*
a. Weight percentage b. Molality
c. Molarity d. Mole fraction

4. What is the mole fraction of the solute in a 1.00 m aqueous solution? *(2015 Re)*
a. 0.0177 b. 0.177 c. 1.770 d. 0.0354

5. How many grams of concentrated nitric acid solution should be used to prepare 250 mL of 2.0 M HNO_3? The concentrated acid is 70% HNO_3. *(2013)*
a. 70.0 g conc. HNO_3 b. 54.0 g conc. HNO_3
c. 45.0 g conc. HNO_3 d. 90.0 g conc. HNO_3

Solubility

6. The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order: *(2024)*
a. A > C > B b. A > B > C
c. B > A > C d. B > C > A

7. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R: *(2023)*
Assertion A: Helium is used to dilute oxygen in diving apparatus.
Reasons R: Helium has high solubility in O_2
In the light of the above statements, choose the correct answer from the options given below:
a. A is false but R is true.
b. Both A and R are true and R is the correct explanation of A.
c. Both A and R are true and R is NOT the correct explanation of A.
d. A is true but R is false.

8. K_H value for some gases at the same temperature 'T' are given: *(2022 Re)*

gas	K_H/k bar
Ar	40.3
CO_2	1.67
HCHO	1.83×10^{-5}
CH_4	0.413

where K_H is Henry's Law constant in water. The order of their solubility in water is:
a. HCHO < CH_4 < CO_2 < Ar
b. Ar < CO_2 < CH_4 < HCHO
c. Ar < CO_2 < CH_4 < HCHO
d. HCHO < CO_2 < CH_4 < Ar

Vapour Pressure of Liquid Solutions

9. Which one is not correct mathematical equation for Dalton's Law of partial pressure? Here p = total pressure of gaseous mixture *(2022)*
a. $p_i = x_i p_i^o$, where x_i = mole fraction of ith gas in gaseous mixture
p_i^o = pressure of ith gas in pure state
b. $p = p_1 + p_2 + p_3$
c. $p = n_1\frac{RT}{V} + n_2\frac{RT}{V} + n_3\frac{RT}{V}$
d. $p_i = x_i p$, where p_i = partial pressure of ith gas
x_i = mole fraction of ith gas in gaseous mixture

10. The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is: *(2021)*
[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]
a. 168 mm of Hg b. 336 mm of Hg
c. 350 mm of Hg d. 160 mm of Hg

11. Which of the following statements about the composition of the vapour over an ideal 1 : 1 molar mixture of benzene and toluene is correct? Assume that the temperature is at 25°C. (Given, vapour pressure data at 25°C, benzene = 12.8 kPa, toluene = 3.85 kPa) *(2016-I)*
a. The vapour will contain equal amounts of benzene and toluene
b. Not enough information is given to make a prediction
c. The vapour will contain a higher percentage of benzene
d. The vapour will contain a higher percentage of toluene

Ideal and Non-Ideal Solutions

12. The mixture which shows positive deviation from Raoult's law is: *(2020)*

a. Benzene + Toluene
b. Acetone + Chloroform
c. Chloroethane + Bromoethane
d. Ethanol + Acetone

13. For an ideal solution, the correct option is : *(2019)*

a. Δ_{mix} S = 0 at constant T and P
b. Δ_{mix} V $\neq$ 0 at constant T and P
c. Δ_{mix} H = 0 at constant T and P
d. Δ_{mix} G = 0 at constant T and P

14. The mixture that forms maximum boiling azeotrope is: *(2019)*

a. Water + Nitric acid
b. Ethanol + Water
c. Acetone + Carbon disulphide
d. Heptane + Octane

15. Which one of the following is incorrect for ideal solution? *(2016 - II)*

a. $\Delta P = P_{obs} - P_{\text{calculated by Raoult's law}} = 0$
b. $\Delta G_{mix} = 0$
c. $\Delta H_{mix} = 0$
d. $\Delta U_{mix} = 0$

16. Which one is not equal to zero for an ideal solution? *(2015)*

a. ΔS_{mix} b. ΔV_{mix}
c. $\Delta P = P_{observed} - P_{Raoult}$ d. ΔH_{mix}

Colligative Properties and Determination of Molar Mass

17. Mass of glucose ($C_6H_{12}O_6$) required to be dissolved to prepare one litre of its solution which is isotonic with 15 g L^{-1} solution of urea (NH_2CONH_2) is (Given: Molar mass in g mol^{-1} C : 12, H : 1, O : 16, N : 14) *(2024 Re)*

a. 55 g b. 15 g c. 30 g d. 45 g

18. The plot of osmotic pressure (P) vs concentration (mol L^{-1}) for a solution gives a straight line with slope 25.73 L bar mol^{-1}. The temperature at which the osmotic pressure measurement is done is: *(2024)*

(Use R = 0.083 L bar mol^{-1} K^{-1})

a. 25.73°C b. 12.05°C
c. 37°C d. 310°C

19. Which amongst the following aqueous solution of electrolytes will have minimum elevation in boiling point? Choose the **correct** option : *(2023-Manipur)*

a. 0.05 M NaCl b. 0.1 M KCl
c. 0.1 M $MgSO_4$ d. 1 M NaCl

20. The following solutions were prepared by dissolving 10 g of glucose ($C_6H_{12}O_6$) in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2) and 10 g of sucrose ($C_{12}H_{22}O_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is: *(2021)*

a. $P_1 > P_2 > P_3$ b. $P_2 > P_3 > P_1$
c. $P_3 > P_1 > P_2$ d. $P_2 > P_1 > P_3$

21. If 8 g of a non-electrolyte solute is dissolved in 114 g of n-octane to reduce its vapour pressure to 80%, the molar mass (in g mol^{-1}) of the solute is

[Given that molar mass of n-octane is 114 g mol^{-1}] *(2020-Covid)*

a. 60 b. 80 c. 20 d. 40

22. Isotonic solutions have same *(2020-Covid)*

a. Freezing temperature b. Osmotic pressure
c. Boiling temperature d. Vapour pressure

23. If molality of the dilute solution is doubled, the value of molal depression constant (K_f) will be: *(2017-Delhi)*

a. Unchanged b. Doubled
c. Halved d. Tripled

24. At 100°C, the vapour pressure of a solution of 6.5 g of a solute in 100 g water is 732 mm. If K_b = 0.52, the boiling point of this solution will be: *(2016 - I)*

a. 103° C b. 101° C c. 100° C d. 102° C

Abnormal Molar Mass

25. The freezing point depression constant (K_f) of benzene is 5.12 K kg mol^{-1}. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places) : *(2020)*

a. 0.80 K b. 0.40 K c. 0.60 K d. 0.20 K

26. The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is: *(2016 - II)*

a. 2 b. 3 c. 0 d. 1

27. Which one of the following electrolytes has the same value of van't Hoff's factor (*i*) as that of $Al_2(SO_4)_3$ (if all are 100% ionised)? *(2015)*

a. $K_3[Fe(CN)_6]$ b. $Al(NO_3)_3$
c. $K_4[Fe(CN)_6]$ d. K_2SO_4

28. The boiling point of 0.2 mol kg^{-1} solution of X in water is greater than equimolal solution of Y in water. Which one of the following statements is true in this case? *(2015)*

a. Molecular mass of X is greater than the molecular mass of Y
b. Molecular mass of X is less than the molecular mass of Y
c. Y is undergoing dissociation in water while X undergoes no change
d. X is undergoing dissociation in water

29. Of the following 0.10 m aqueous solutions, which one will exhibit the largest freezing point depression? *(2014)*

a. $C_6H_{12}O_6$ b. $Al_2(SO_4)_3$ c. K_2SO_4 d. KCl

Answer Key

1. (a)	**2.** (c)	**3.** (c)	**4.** (a)	**5.** (c)	**6.** (d)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (b)
11. (c)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (a)	**17.** (d)	**18.** (c)	**19.** (a)	**20.** (d)
21. (d)	**22.** (b)	**23.** (a)	**24.** (b)	**25.** (b)	**26.** (b)	**27.** (c)	**28.** (d)	**29.** (b)	